Hegel Cognitive Architecture First Reconstruction

Research note • 2 November 2026 • First calibration packet • Provisional interpretation

Companion to the research protocol for deriving a general cognitive architecture from a single author's corpus, with Hegel as the first case and Hegel in Delhi as a possible later testbed.

Finding

The inspected passages support a family of architectural constraints: interdependent mental activities, acquired habits that alter conscious engagement, different operations for images and names, integrated knowing and willing, and socially constituted self-relations. A further, more demanding interpretation permits revision of the categories and standards through which experience is judged.

This is a substantive starting point for architecture derivation. It does not uniquely determine a runnable architecture. The target is a general account of cognition articulated by the corpus. Its applicability across simulated entities and its empirical adequacy remain separate questions.

The principal uncertainty is now precise: which relations in a philosophical account can become computational dependencies without changing their meaning?

Evidence boundaries

The calibration centers on selected passages of the 1830 Encyclopaedia, supplemented by the 1807 Phenomenology introduction and the Philosophy of Right introduction. This is purposive sampling, not a completed extraction of Hegel's corpus. The works differ in date, purpose, and argumentative level; cross-work agreement is corroboration within one author, not independent empirical evidence.

German and English witnesses were compared for the consequential Encyclopaedia relations and the Phenomenology's account of experience. The Philosophy of Right check below uses the English witness only and is supplementary. No critical-edition or manuscript collation has been completed.

Wallace's 1894 translation states that it omits the editors' additions. Its introductory essays are Wallace's. The German text with additions marks passages as "Zusatz"; these are kept separate from numbered main text and authored remarks. The short German transcription omits some remarks, so absence there cannot establish absence from Hegel. Section numbers provide the alignment key.

Every passage inspected for this packet is development material. No held-out fidelity score is reported. An untouched evaluation sample must be frozen before the next extraction round. "Untouched" refers to this workflow; it does not establish that an LLM has never encountered the text during training.

Claim ledger

Each record separates a textual relation from a proposed architectural constraint. The constraint and its implementation are our hypotheses. Relation types distinguish constitutive, conceptual, normative, developmental, and enabling claims from empirical causal laws.

The challenge column identifies either counterevidence, a competing reading, or a missing specification. None of the records has yet received independent review by a Hegel specialist.

Scope and organization

ID and source	Textual relation and type	Proposed constraint	Challenge and unknowns
H01 · Encyclopaedia §377, German / English	The inquiry concerns spirit generally, beyond individual traits. **Scope; authored claim.**	Extract forms of cognition separately from biography and character content.	Generality is a philosophical aim; population-wide validity does not follow.
H02 · §§379-380 and §445 remark, German / English	Activities belong to an integrated whole; lower determinations persist as moments of developed spirit. **Constitutive.**	Specify how activities depend on and modify one another. A list of isolated faculties is inadequate.	Integration does not require one software module, shared storage, or a particular network topology.
H03 · §442 remark, German / English	Conceptual development is distinguished from the individual's successive acquisition of faculties. **Conceptual order; explicit qualification.**	Keep conceptual dependency separate from runtime order and learning chronology.	The passage supplies no scheduler, activation threshold, concurrency rule, or clock.

Mental activity

ID and source	Textual relation and type	Proposed constraint	Challenge and unknowns
H04 · §§409-410, German / English	Practice produces habit; habitual activity leaves engagement available elsewhere. **Developmental and enabling.**	Habit acquisition should affect how an activity occupies cognition.	Habit can also constrain freedom or conflict with another purpose. Learning rates, transfer, and interruption rules are unspecified.
H05 · §§447-448, German / English	Attention is active; cultivated feeling involves acquired distinctions. **Constitutive and developmental.**	Test whether learned distinctions alter what becomes salient or discriminable.	A feedback path from concepts to noticing is an interpretation. Quantitative attention capacity is not supplied.
H06 · §§452-453, German / English	An inward image differs from immediate intuition and can persist outside consciousness. **Representational distinction.**	Distinguish an event's occurrence, retained content, and current availability.	No storage format, decay function, or biological memory mechanism follows.
H07 · §454, German / English	Present intuition and retained image enter a relation of reidentification. **Conceptual transformation.**	Make recognition of something previously encountered distinguishable from merely recording it.	A similarity metric or cue-triggered retrieval routine would be an engineering addition.
H08 · §§461-464, German / English	Name-based remembering can operate without an image; memory has a relation to thinking. **Representational and constitutive.**	Preserve a functional distinction between image-mediated and name-mediated access.	This does not establish separate databases, modern episodic/semantic stores, or a mandatory serial pipeline.
H09 · §468, German / English	Intelligence determining its content is characterized as will. **Conceptual transition.**	Explain how interpretation and practical determination remain connected.	Sharing a record between a planner and a reasoner is only one implementation; utility functions are not given.

An authored illustration sharpens H08: §462 uses understanding the name of a lion without picturing the animal. This motivates distinguishing name-based access from image reconstruction. It is an explanatory example, not experimental evidence for separate memory stores.

Social relations and revision

ID and source	Textual relation and type	Proposed constraint	Challenge and unknowns
H10 · Encyclopaedia §436, German / English	Universal self-consciousness involves reciprocal recognition of free others. **Social and constitutive.**	Represent recognition as a relation involving both agents and their self-understanding.	Reducing this to approval, reputation, or trust loses content. A transition rule for everyday encounters is not specified.
H11 · Phenomenology introduction §§84-86, German, Werke 3 pp76-78 / English	Examination can alter knowledge, its object, and its criterion. **Epistemic transformation.**	Permit revision of the agent's framing and standards, beyond confidence changes within a fixed framing.	Transposing this account of consciousness into an ordinary decision mechanism needs argument. It does not authorize changing the simulation's external facts.
H12 · Introduction §87, German, Werke 3 p79 / English	The observer's reconstruction of the transition is not wholly available to the consciousness undergoing it. **Explanatory standpoint.**	Keep evaluator knowledge separate from what an entity can access or report.	A verbose self-explanation is not evidence that the represented process occurred.
H13 · Philosophy of Right §§5-7, English	Abstraction from a determination and commitment to a particular end are moments of willing. **Conceptual and normative.**	Include both commitment and the possibility of reconsidering it.	This is an account of freedom, not an empirically calibrated choice rule. German alignment remains pending.

Rejected position as a control

H14 · Phenomenology introduction §§73-76. The opening discussion considers, then criticizes, treating knowledge as an external instrument or medium. It must not be coded as Hegel endorsing that model. Relation type: **reported position under criticism**. Source: English introduction, cross-checked against the German introduction.

An extraction that turns every affirmative-looking sentence into an architectural rule fails this control. The same attribution check applies to opponents, examples, imagined speakers, and editorial additions throughout the corpus.

Translation decisions that affect the model

Source term	Working treatment	Consequence
Gewohnheit	Acquired habit, including embodied facility	A cached answer alone does not capture H04.
Aufmerksamkeit	Active attention	Calling an LLM attention mechanism by this name establishes no correspondence.
Erinnerung and Gedächtnis	Preserve the distinction between inward retention/recollection and the later discussion of names and memory	One undifferentiated retrieval store would need an argument for how it preserves the distinct functions.
Vorstellung and Denken	Keep representation and conceptual thinking distinguishable	A text embedding does not, by itself, implement the relation between them.
Anerkennung	Reciprocal recognition in the relevant social argument	Agreement with a proposition and recognition of another agent must remain distinguishable.
Erfahrung	The specific transformation discussed in §§84-87	Updating a numerical confidence is only one possible kind of learning and does not exhaust H11.

These are operational translation decisions for review, not a new glossary claiming one-to-one equivalence with contemporary cognitive science.

A provisional architecture specification

The following organization is our synthesis of the ledger. It is not Hegel’s own software specification.

Architectural relation	Evidence	Required observable difference if implemented
Acquired routines alter the demands of ongoing activity	H04-H05	Different histories of practice change execution and concurrent noticing.
Retained images, present recognition, and name-based access have distinguishable roles	H06-H08	Image cues and learned names can support different retrieval paths.
Conceptual interpretation participates in practical determination	H02, H09, H13	Changing a relevant interpretation can affect commitments and actions.
Another agent’s recognition bears on self-relation	H10	Reciprocity and unilateral approval can produce different state changes.
A mismatch can provoke revision of framing or criteria	H11	Transfer to a new case can change after revision, beyond repeating a corrected answer.
Entity-accessible content is bounded separately from the evaluator’s reconstruction	H12	Behavior cannot depend on hidden world state or an evaluator’s explanation.

The common architecture would specify the operations and permitted relationships. Each entity would have its own acquired habits, memories, distinctions, commitments, social history, and circumstances. Applying the same architecture therefore need not produce entities with Hegel’s opinions or identical behavior.

Persistent state would minimally distinguish acquired routines, representational contents, name-content relations, practical commitments, social relations, and current criteria. “Distinguish” is functional: the implementation could share storage. Transient state would include currently engaged content and unresolved discrepancies.

No source-supported numerical specification yet exists for memory limits, learning speed, attention allocation, discrepancy thresholds, or stopping conditions. The world interface, scheduler, logging, and compute budget must be supplied explicitly as engineering assumptions.

Two formalization families

Both candidates must preserve the core source distinctions. Neither is an independent-faculties straw man.

Question	A · Fixed operations with revisable representations	B · Revisable organization of operations
What is fixed?	Operator repertoire and the allowed dependency structure.	A bounded language of operators and permitted organizational revisions.
What changes?	Representations, criteria expressed in the existing language, routine parameters, commitments, and social relations.	The same contents, plus eligible dependencies or applicability rules connecting operations.
Example	A learned category changes interpretation through a pre-existing feedback path.	Learning establishes or restructures which distinctions gate attention, recognition, or practical deliberation.
Relation to the text	A conservative computational realization of integration and transformation.	A stronger hypothesis about implementing transformation.
Main risk	A fixed vocabulary can prevent changes that the reconstruction requires.	Extra flexibility can improve fit without being licensed by the corpus.
What would distinguish them?	Its reachable state transitions must be specified before testing.	A proposed revision must produce an effective dependency unavailable to A under the agreed representation language.

There is a real equivalence problem. A fixed interpreter can represent another program as data. Consequently, rewriting code versus updating a JSON record does not by itself distinguish cognitive architectures. Compare effective dependencies and counterfactual behavior under explicit resource limits. If the candidates simulate one another within those limits, merge them or revise the comparison.

Corti’s study of Hegel’s later cognition argues for a transformative reading in which higher capacities affect lower activities. It also presents this as a contested interpretive position. That supports examining reciprocal influence, while leaving the choice between A and B open. Philosophical transformation does not entail runtime graph rewriting.

How the arXiv methods help

Study and inspected version	Relevant result	Use in this project and limit
Rmus et al. - GeCCo, v3, 7 November 2025	Generates and fits executable cognitive models across four task domains. Its methods separate prompt data, validation feedback, and a third test set.	Search over explicit candidate mechanisms after the textual constraints are established. Log what the template supplies. Task-level model discovery does not establish whole-architecture induction from an author.
Xie et al. - Think-Aloud Reshapes Automated Cognitive Model Discovery Beyond Behavior, v1, 6 May 2026	Adding think-aloud reports changes the structural families of discovered models in a risky-choice task. Programs are compared through normalized computation graphs.	Test whether corpus evidence changes model organization, beyond wording or fitted parameters. A philosophical corpus is not a participant's concurrent verbal report. Trials used to select the best model must not be described as an untouched final test.
Wurgaft et al. - Scaling up the think-aloud method, v1, 29 May 2025	Converts verbal protocols into search graphs, with moderate agreement against human coding.	Validate extraction against independently annotated passages and inspect disagreement. Its task-specific graph vocabulary is supplied rather than discovered from an author.
Park et al. - Generative Agents, v2, 6 August 2023	Evaluates a memory, reflection, and planning architecture and component ablations for believable simulation behavior.	Use as a simulation comparator and an example of ablation-based evaluation. Believability alone does not establish textual fidelity or human cognitive accuracy.

The research supports components of the method. The proposed extension is to use an author's process claims to constrain a general architecture, then test its implementations. None of these inspected studies establishes that complete extension.

To test whether derivation actually uses the corpus, compare extraction with the passages, with author identity alone, and without either. Perturb or remove specific supporting passages and check whether the corresponding proposed dependencies change. These controls can expose unsupported prior assumptions; they cannot prove that pretrained knowledge is absent.

Concrete tests for Hegel in Delhi

These are designed experiments, not results. Each test needs a frozen specification, matched information and compute, and traces of the relevant state changes.

Test	Intervention	Measurements and failure signal
Habit and attention	Compare a practiced route with a novel route; introduce an incidental event and later a changed route.	Route execution, event noticing and recall, and adaptation. Faster execution alone is insufficient evidence for the proposed attention relation.
Image and name access	Supply an image-like cue, a learned name, or a new label for a familiar object.	Recognition, retrieval errors, and transfer between cues. Disable one access operation to test its causal role.
Revision of a category	Create cases in which the entity's current interpretation produces systematic failure, then present a novel case.	Whether the representation or criterion changes; whether that change predicts transfer. More persuasive explanations without changed behavior fail.
Reciprocal recognition	Hold praise and material reward constant while changing whether recognition is reciprocal or unilateral.	Commitment, self-description, and treatment of the other entity. A change driven only by a sentiment score is inadequate for H10.
Knowing and willing	Change a relevant interpretation while keeping available actions and resources fixed.	Whether practical commitments and subsequent actions respond coherently. A rewritten narrative with unchanged decision dependencies fails.
Limits of self-knowledge	Keep a world fact available to the evaluator but outside the entity's observations.	Check that decisions do not use it. Compare the entity's explanation with its recorded operations rather than treating the explanation as ground truth.

Compare the current simulation, a generic architecture given the same relevant corpus material, and the derived candidates. Use component ablations that keep access to information constant where possible. Shared exogenous events may be controlled; consequences of different actions must remain free to diverge.

Keep three judgments separate: **textual fidelity**, **empirical plausibility**, and **simulation utility**. Human behavior or independently justified cognitive findings are needed for the second. An LLM judge's preference for Hegel-like prose cannot substitute for any of the three.

Status and next research gate

This packet establishes fourteen attributable records, a provisional set of organizational constraints, two implementation families, and six discriminating test designs. It is a first reconstruction, not a validated general theory or a measured improvement to the simulation. No code, training, or simulation run was performed.

Before scaling extraction, the immediate gate is an independent source-competent review of H02, H05, and H11, where conceptual integration, cultivated perception, and criterion revision risk being over-translated into causal machinery. Freeze a separate textual challenge sample before that next round, preserving parallel translations as one evidence unit.

Then decide whether A and B can be made behaviorally distinct under a bounded specification. If they cannot, keep the shared constraints and stop presenting them as different architectures. A successful derivation must survive counterexamples and produce consequential organizational commitments; otherwise it remains a thematic interpretation.